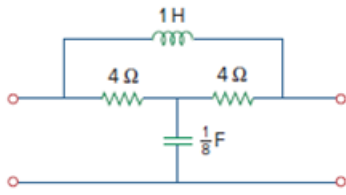


Problem

Obtain the h parameters at $\omega = 4$ rad/s for the circuit in Fig. 19.124 using *PSpice* or *MultiSim*.

Figure 19.124



Step-by-step solution

Step 1 of 9

Draw the circuit diagram as shown below.

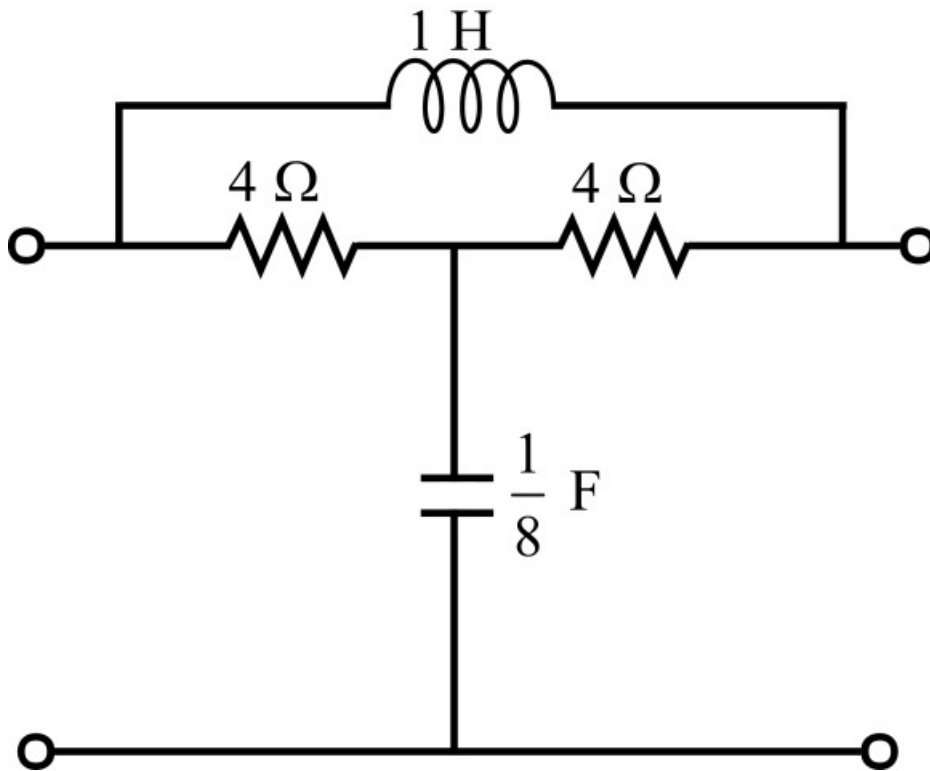


Figure 1

Step 2 of 9

Re-draw Figure 1, with node parameters.

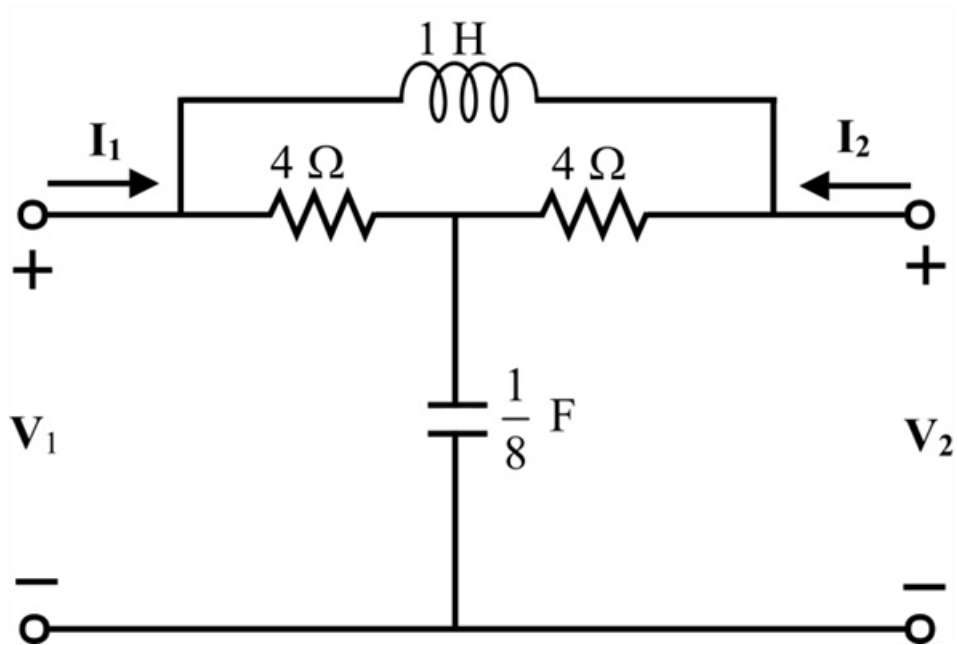


Figure 2

Step 3 of 9

Consider, $\omega = 4 \text{ rad/s}$.

$$\begin{aligned} f &= \frac{\omega}{2\pi} \\ &= \frac{4}{2\pi} \\ &= 0.6366 \text{ Hz} \end{aligned}$$

Calculate h -parameters.

$$\begin{aligned} h_{11} &= \left. \frac{V_1}{I_1} \right|_{V_2=0} & h_{12} &= \left. \frac{V_1}{V_2} \right|_{I_1=0} \\ h_{21} &= \left. \frac{I_2}{I_1} \right|_{V_2=0} & h_{22} &= \left. \frac{I_2}{V_2} \right|_{I_1=0} \end{aligned}$$

Step 4 of 9

Calculate h_{11} and h_{21} .

(i) Set $V_2 = 0$, $I_1 = 1$ A, start freq = 0.6366 Hz and End freq = 0.6366 Hz.

Draw Pspice to obtain h_{11} and h_{21} .

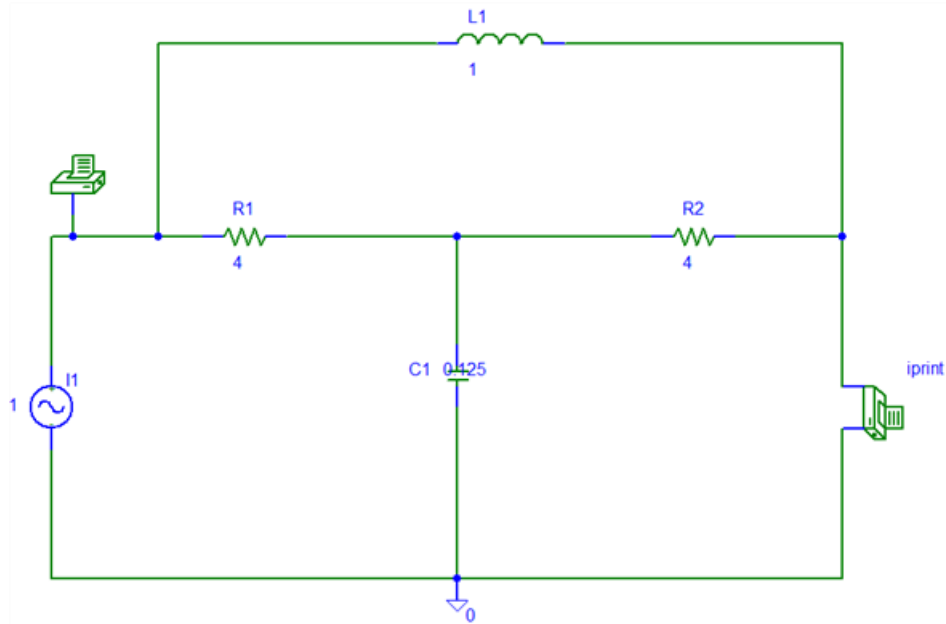


Figure 3

Step 5 of 9

From Figure 3, the following Pspice results are obtained.

FREQ IM(V_PRINT2) IP(V_PRINT2)

6.363E-01 1.202E+00 1.463E+02

FREQ VM(\$N_0002) VP(\$N_0002)

6.363E-01 3.770E+00 -1.350E+02

From the results of Pspice, Calculate h_{11} and h_{21} values.

Calculate h_{11} value.

$$\begin{aligned} h_{11} &= \frac{V_1}{I_1} \\ &= \frac{3.771 \angle -135^\circ}{1} \\ &= 3.771 \angle -135^\circ \end{aligned}$$

Therefore, the value of h_{11} is $3.771 \angle -135^\circ$.

Calculate h_{21} value.

$$\begin{aligned} h_{21} &= \frac{I_2}{V_2} \\ &= \frac{1.202 \angle 146.3^\circ}{1} \\ &= 1.202 \angle 146.3^\circ \end{aligned}$$

Therefore, the value of h_{21} is $1.202 \angle 146.3^\circ$.

Step 6 of 9

Calculate h_{12} and h_{22} .

(ii) Set $I_1 = 0$, $V_2 = 1$ V, start freq = 0.6366 Hz and End freq = 0.6366 Hz.

Draw Pspice to obtain h_{12} and h_{22} .

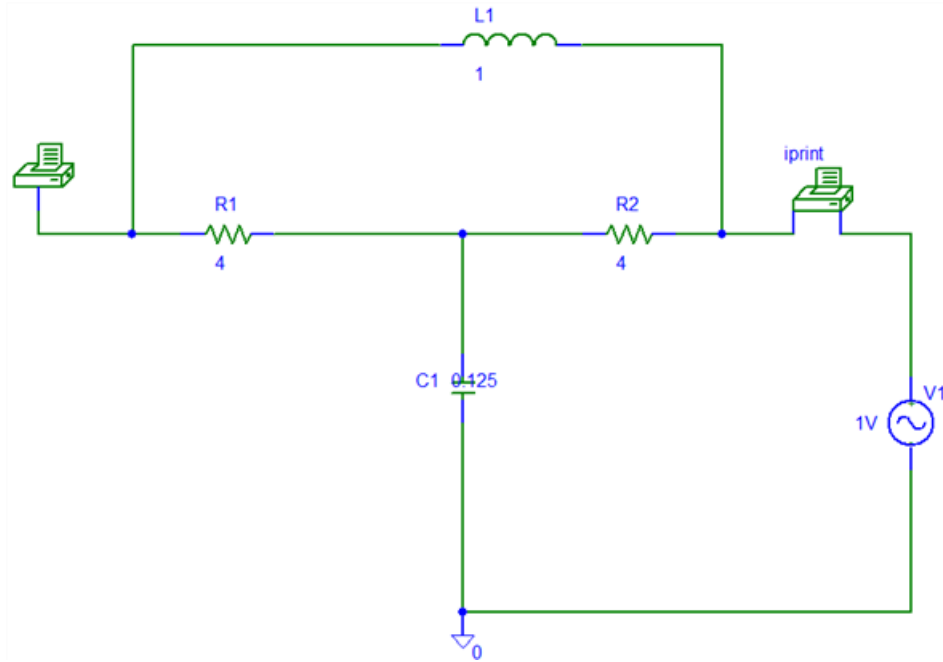


Figure 4

Step 7 of 9

From Figure 4, the following Pspice results are obtained.

FREQ VM(\$N_0002) VP(\$N_0002)

6.363E-01 1.202E+00 -3.369E+01

FREQ IM(V_PRINT2) IP(V_PRINT2)

6.363E-01 3.726E-01 -1.534E+02

Step 8 of 9

From the results of Pspice, Calculate h_{12} and h_{22} values.

Calculate h_{12} value.

$$\begin{aligned} h_{12} &= \frac{V_1}{V_2} \\ &= \frac{1.202 \angle -33.69^\circ}{1} \\ &= 1.202 \angle -33.69^\circ \end{aligned}$$

Therefore, the value of h_{12} is $1.202 \angle -33.69^\circ$.

Calculate h_{22} value.

$$\begin{aligned} h_{22} &= \frac{I_2}{V_2} \\ &= \frac{0.3727 \angle -153.4^\circ}{1} \\ &= 0.3727 \angle -153.4^\circ \end{aligned}$$

Therefore, the value of h_{22} is $0.3727 \angle -153.4^\circ$.

Step 9 of 9

Therefore, the h -parameters are:

$$[h] = \begin{bmatrix} 3.771 \angle -135^\circ & 1.202 \angle -33.69^\circ \\ 1.202 \angle 146.3^\circ & 0.3727 \angle -153.4^\circ \end{bmatrix}$$